

TRAFFICSENTINEL VLM

Standalone Traffic Violation Detection & Cryptographic Court Evidence Locker

Project Submission & Technical Documentation

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Target: AIS Evaluation Suite

1. Executive Summary

TrafficSentinel VLM is an intelligent edge-ready computer vision solution designed to detect critical traffic violations, recognize license registration details via ANPR OCR, and manage secure judicial evidence packages. The project integrates natural language vision queries with vision-language models (VLM), a human-in-the-loop judicial verification system, cryptographic evidence signing, and high-performance hardware-based image preprocessing presets.

System Architecture

The solution is built using a decoupled architecture, allowing both components to run on separate hosts or containers:

- **FastAPI Backend API:** Serves model inference simulations, calculates evidence packet checksums using HMAC SHA-256, and handles custom prompt inputs.
- **Vanilla HTML/CSS/JS Dashboard:** A premium web application showcasing camera locations, video player simulations with dynamic canvases for bounding boxes, an officer review queue, and a theme switcher supporting Dark Mode.

2. System Features & Specifications

Interactive Preset Scenario Suite

The platform comes preloaded with 7 scenarios covering key road safety violations:

- **Scenario 1 (Two-Wheeler):** Helmet non-compliance check.
- **Scenario 2 (Overcrowding):** Triple riding on a single scooter.
- **Scenario 3 (Intersection):** Red light violation crossing a stop line.
- **Scenario 4 (Obstruction):** Illegal parking on a sidewalk.
- **Scenario 5 (Danger Flow):** Driving against traffic direction.
- **Scenario 6 (Windshield):** Driver seatbelt non-compliance.
- **Scenario 7 (Encroachment):** Vehicle stopped past the painted line.

Zero-Shot VLM Prompter & OCR

Powered by simulated VLM-LLaVA-v3, the prompter allows raw natural language queries to search the visual frame. Simultaneously, the ANPR OCR module identifies and extracts license plate strings, returning confidence ratings and visualizing results using real HSRP plate mockups.

Cryptographic Evidence Locker & Judicial Audit

A critical feature for judicial admissibility is the **Evidence Locker**. Every infraction creates a secure JSON payload signed with a SHA-256 HMAC hash. To ensure human verification, a **Judicial Review Queue** is integrated. If a human operator rejects the infraction, the status transitions to 'Rejected' (False Positive) which immediately invalidates the cryptographic lock, rendering the signature VOID in the evidence file. This guarantees audit integrity.

Low-Light Preprocessing Pipeline

Includes hardware contrast presets (CLAHE, gamma adjustment) designed to clarify low-visibility night feeds before feeding them into the VLM.

Premium Dark Mode Integration

A theme-toggle button leverages modern CSS variables, transitioning background states, borders, active gallery cards, and buttons seamlessly, storing preferences locally via localStorage to prevent any white-screen flicker during boot.

3. Installation & Run Instructions

Step 1: Set Up Virtual Environment

Create and activate a python virtual environment:

```
# Create environment
python -m venv .venv

# Activate environment (Windows)
.venv\Scripts\activate

# Activate environment (macOS/Linux)
source .venv/bin/activate
```

Step 2: Install Dependencies

```
pip install fastapi uvicorn pydantic
```

Step 3: Run the Backend API

Navigate to the root directory and start the FastAPI server on port 8002:
python api/main.py

Step 4: Run the Frontend Server

In a separate terminal window, start a lightweight web server inside the dashboard directory:
cd dashboard
python -m http.server 3002

Step 5: Load the Application & Verify

Open a web browser and navigate to **http://localhost:3002**. Test presets by switching scenario cards, writing custom zero-shot prompter phrases, toggling Dark Mode, and exporting cryptographic signed dossiers using the download action.

Table 1. TrafficSentinel VLM Component Grid

Component	Functionality	Default Port
FastAPI Server	VLM & ANPR OCR Simulation Engine	8002 (HTTP)
Static Web Server	HTML5/CSS3/JS Interactive Dashboard	3002 (HTTP)
Evidence Locker	HMAC SHA-256 cryptographic signatures	N/A